

1. The induced emf can be produced in a coil by
- moving the coil with uniform speed inside uniform magnetic field
 - moving the coil with non uniform speed inside uniform magnetic field
 - rotating the coil inside the uniform magnetic field
 - changing the area of the coil inside the uniform magnetic field

Choose the correct answer from the options given below:

[2023 (06 Apr Shift 1)]

- B and C only
- A and C only
- C and D only
- B and D only

2. For the plane electromagnetic wave given by $E = E_0 \sin(\omega t - kx)$ and $B = B_0 \sin(\omega t - kx)$, the ratio of average electric energy density to average magnetic energy density is

[2023 (06 Apr Shift 1)]

- $\frac{1}{2}$
- 2
- 4
- 1

3. Two concentric circular coils with radii 1 cm and 1000 cm and number of turns 10 and 200 respectively are placed coaxially with centers coinciding. The mutual inductance of this arrangement will be $\times 10^{-8}$ H. (Take, $\pi^2 = 10$)

[2023 (06 Apr Shift 2)]

4. An emf of 0.08 V is induced in a metal rod of length 10 cm held normal to a uniform magnetic field of 0.4 T, when move with a velocity of:

[2023 (08 Apr Shift 2)]

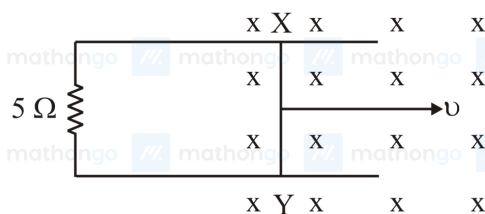
- 0.5 m s^{-1}
- 20 m s^{-1}
- 3.2 m s^{-1}
- 2 m s^{-1}

5. The energy of an electromagnetic wave contained in a small volume oscillates with

[2023 (10 Apr Shift 1)]

- Double the frequency of the wave
- The frequency of the wave
- Half the frequency of the wave
- Zero frequency

6. A 1 m long metal rod XY completes the circuit as shown in figure. The plane of the circuit is perpendicular to the magnetic field of flux density 0.15 T. If the resistance of the circuit is 5Ω , the force needed to move the rod in direction, as indicated, with a constant speed of 4 m s^{-1} will be $\times 10^{-3}$ N.



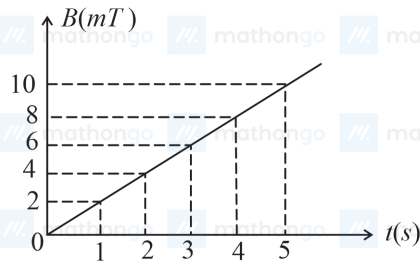
[2023 (10 Apr Shift 1)]

7. A square loop of side 2.0 cm is placed inside a long solenoid that has 50 turns per centimetre and carries a sinusoidally varying current of amplitude 2.5 A and angular frequency 700 rad s^{-1} . The central axes of the loop and solenoid coincide. The amplitude of the emf induced in the loop is $x \times 10^{-4}$ V. The value of x is

(Take, $\pi = \frac{22}{7}$)

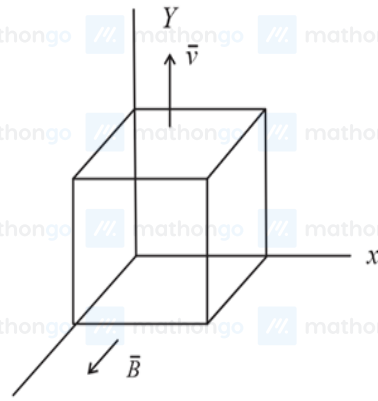
[2023 (10 Apr Shift 2)]

8. The magnetic field B crossing normally a square metallic plate of area 4 m^2 is changing with time as shown in figure. The magnitude of induced emf in the plate during $t = 2 \text{ s}$ to $t = 4 \text{ s}$, is _____ mV .



[2023 (11 Apr Shift 1)]

9. A metallic cube of side 15 cm moving along y -axis at a uniform velocity of 2 m s^{-1} . In a region of uniform magnetic field of magnitude 0.5 T directed along z -axis. In equilibrium the potential difference between the faces of higher and lower potential developed because of the motion through the field will be mV .



[2023 (11 Apr Shift 2)]

10. A coil has an inductance of 2 H and resistance of 4Ω . A 10 V is applied across the coil. The energy stored in the magnetic field after the current has built up to its equilibrium value will be _____ $\times 10^{-2} \text{ J}$

[2023 (11 Apr Shift 2)]

11. A conducting circular loop is placed in a uniform magnetic field of 0.4 T with its plane perpendicular to the field. Somehow, the radius of the loop starts expanding at a constant rate of 1 mm s^{-1} . The magnitude of induced emf in the loop at an instant when the radius of the loop is 2 cm will be _____ μV .

[2023 (12 Apr Shift 1)]

12. The source of time varying magnetic field may be

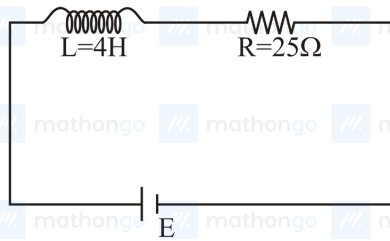
- (A) a permanent magnet
- (B) an electric field changing linearly with time
- (C) direct current
- (D) a decelerating charge particle
- (E) an antenna fed with a digital signal

Choose the correct answer from the options given below.

[2023 (13 Apr Shift 1)]

- (1) (C) and (E) only
- (2) (D) only
- (3) (A) only
- (4) (B) and (D) only

13. In the given figure, an inductor and resistor are connected in series with a battery of emf E volt. $\frac{E^2 a}{2b}$ J s⁻¹ represents the maximum rate at which the energy is stored in the magnetic field (inductor). The numerical value of $\frac{b}{a}$ will be _____.



[2023 (13 Apr Shift 1)]

14. An insulated copper wire of 100 turns is wrapped around a wooden cylindrical core of the cross-sectional area 24 cm^2 . The two ends of the wire are connected to a resistor. The total resistance in the circuit is 12Ω . If an externally applied uniform magnetic field in the core along its axis changes from 1.5 T in one direction to 1.5 T in the opposite direction, the charge flowing through a point in the circuit during the change of magnetic field will be _____ mC.

[2023 (13 Apr Shift 2)]

15. A 12 V battery connected to a coil of resistance 6Ω through a switch, drives a constant current in the circuit. The switch is opened in 1 ms . The emf induced across the coil is 20 V . The inductance of the coil is :

[2023 (15 Apr Shift 1)]

- (1) 10 mH
- (2) 8 mH
- (3) 5 mH
- (4) 12 mH

16. A 20 cm long metallic rod is rotated with 210 rpm about an axis normal to the rod passing through its one end. The other end of the rod is in contact with a circular metallic ring. A constant and uniform magnetic field 0.2 T parallel to the axis exists everywhere. The emf developed between the centre and the ring is _____ mV.

(Take $\pi = \frac{22}{7}$)

[2023 (15 Apr Shift 1)]

ANSWER KEYS

1. (3) 2. (4) 3. (4) 4. (4) 5. (1) 6. (18) 7. (44) 8. (8)
9. (150) 10. (625) 11. (50) 12. (2) 13. (25) 14. (60) 15. (1) 16. (88)

1. (3)

Induced emf is given by Faraday's law

$$|\varepsilon| = \frac{d\phi}{dt}$$

where ϕ is the magnetic flux.

The rate of change of flux through a circuit is defined as emf. The formula for flux is $\phi = \vec{B} \cdot \vec{A}$. In the statements A & B given in the question, the magnetic flux in the coil in uniform magnetic field does not change. So, no emf is induced.

But in statement C, the angle between the area vector and the magnetic field continuously changes hence, an emf is generated. In statement D, the area is changing in unit time so magnetic flux will change and an emf is generated.

2. (4)

The formula to calculate the average electric energy density is given by

$$U_E = \frac{1}{2} \varepsilon_0 E^2 \quad \dots (1)$$

and, the formula to calculate average magnetic energy density is given by

$$U_B = \frac{1}{2\mu_0} B^2 \quad \dots (2)$$

For any particular electromagnetic wave, both magnetic and electric field are equally involved in contributing to energy density. Hence, the ratio of average electric energy density to magnetic energy density is 1.

3. (4)

Let the magnetic field of the 200 turns be B_1 . The formula of magnetic field is given by $B = \frac{\mu_0 i}{2r}$

Let $r = 1000$ cm. Thus,

$$B_1 = \frac{n\mu_0 \times I}{2 \times 10} = \frac{200\mu_0 I}{2 \times 10}$$

The flux through the coil of radius $r_1 = 1$ cm is $\phi = 10 \times (\vec{B} \cdot \vec{A}) = 10 \times B_1 \pi r_1^2$

$$\Rightarrow \phi = 10 \times \frac{200\mu_0 I}{2 \times 10} \times \pi (0.01)^2$$

$$L = 10 \times \frac{200\mu_0}{2 \times 10} \times \pi (0.01)^2$$

$$= 10 \times \frac{200 (4\pi \times 10^{-7})}{2 \times 10} \times \pi (0.01)^2$$

$$= 4 \times 10^{-8} \text{ H}$$

4. (4)

The formula to calculate the motional emf induced in a moving conductor in a magnetic field is given by

$$\varepsilon = Blv \quad \dots (1)$$

Substitute the values of the known parameters into equation (1) and solve to calculate the required velocity of the rod.

$$0.08 \text{ V} = 0.4 \text{ T} \times \left(10 \text{ cm} \times \frac{1 \text{ m}}{100 \text{ cm}}\right) \times v$$

$$\Rightarrow v = \frac{0.08 \text{ V}}{0.4 \text{ T} \times 10 \text{ cm} \times \frac{1 \text{ m}}{100 \text{ cm}}}$$

$$= 2 \text{ m s}^{-1}$$

5. (1)

The formula to calculate the energy confined within the electric component of an electromagnetic wave can be written as

$$u_E = \frac{1}{2} \epsilon_0 E^2 dV \dots (1)$$

Let's assume that the equation of the electric component is given by

$$E = E_0 \sin(\omega t + \phi) \dots (2)$$

Substitute the expression for the electric field component from equation (2) into equation (1) and simplify to obtain the expression for the energy confined within the small volume dV .

$$\begin{aligned} u_E &= \frac{1}{2} \epsilon_0 [E_0 \sin(\omega t + \phi)]^2 dV \\ &= \frac{1}{2} \epsilon_0 E_0^2 \sin^2(\omega t + \phi) dV \\ &= \frac{1}{4} \epsilon_0 E_0^2 [1 - \cos(2\omega t + 2\phi)] \dots (3) \end{aligned}$$

From equation (3), it can be concluded that the frequency of oscillation of the energy confined within a small volume is double the frequency of the wave.

6. (18)

The force on the rod is given by

$$\begin{aligned} F &= i l B \\ \Rightarrow F &= \frac{B l}{R} \times i l B \\ \Rightarrow F &= \frac{B^2 l^2 v}{R} \dots (i) \end{aligned}$$

The data given is

$$R = 5 \Omega$$

$$l = 1 \text{ m}$$

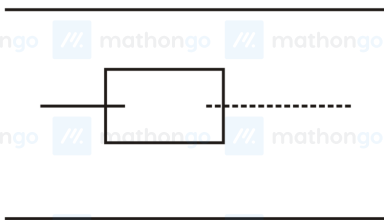
$$v = 4 \text{ m s}^{-1}$$

$$B = 0.15 \text{ T}$$

Substituting the values in equation (i)

$$\begin{aligned} F &= \frac{(0.15)^2 \times (1)^2 \times 4}{5} \\ &= 18 \times 10^{-3} \text{ N} \end{aligned}$$

7. (44)



It is given that the current is varying sinusoidally. The current can be written as $I = I_0 \sin \omega t$.

The magnetic field through the solenoid $B = \mu_0 n I$

The flux through the square is $\phi = \mu_0 n I A$

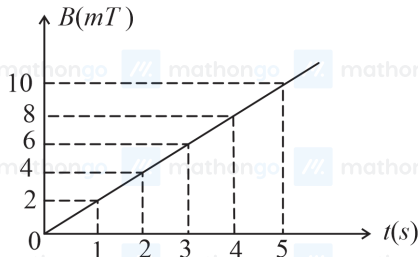
The emf is

$$\begin{aligned} \epsilon &= \mu_0 n A \times \frac{d(I_0 \sin \omega t)}{dt} \\ \Rightarrow \epsilon &= \mu_0 n A I_0 \omega \cos \omega t \end{aligned}$$

The amplitude of the emf is,

$$\begin{aligned} \epsilon &= \mu_0 n A I_0 \omega = 4\pi \times 10^{-7} \times \frac{50}{10^{-2}} \times 4 \times 10^{-4} \times 2.5 \times 700 \\ \Rightarrow \epsilon &= 44 \times 10^{-4} \text{ V} \end{aligned}$$

8. (8)



From the graph, it can be seen that the value of the magnetic field at $t = 2$ s is 4 mT, whereas the magnetic field at $t = 4$ s is 8 mT.

The formula to calculate the magnitude of the induced emf is given by

$$\begin{aligned}\epsilon_{\text{ind}} &= \frac{d\phi}{dt} \\ &= \frac{d}{dt}(BA) \\ &= A \frac{dB}{dt} \dots (1)\end{aligned}$$

$\frac{dB}{dt}$ is the slope of the given graph.

Hence, the value of the induced emf can be calculated as follows:

$$\begin{aligned}\epsilon_{\text{ind}} &= 4 \text{ m}^2 \times \left(\frac{8 \text{ mT} - 4 \text{ mT}}{4 \text{ s} - 2 \text{ s}} \right) \\ &= 8 \text{ mV}\end{aligned}$$

9. (150)

Using the equation

$$\begin{aligned}qE &= qvB \\ \Rightarrow E &= vB\end{aligned}$$

Using the relation $\Delta V = Ed$

$$\begin{aligned}\Rightarrow \Delta V &= vBd \\ &= 2 \times \frac{1}{2} \times 0.15 \text{ V} \\ &= 0.15 \text{ V} = 150 \text{ mV}\end{aligned}$$

10. (625)

The data given is

$$L = 2 \text{ H}$$

$$R = 4 \Omega$$

$$V = 10 \text{ V}$$

Using Ohm's law, $V = IR$,

$$I = \frac{10 \text{ V}}{4 \Omega} = \frac{5}{2} \text{ A}$$

The energy of an inductor is given by

$$E = \frac{1}{2} LI^2 = \frac{1}{2} \times 2 \times \left(\frac{25}{4} \right) = 625 \times 10^{-2} \text{ J}$$

11. (50)

The data given is

$$B = 0.4 \text{ T}$$

$$R = 2 \times 10^{-2} \text{ m}$$

$$\text{It is given that } \frac{dR}{dt} = 10^{-3} \text{ m s}^{-1}$$

Area of the loop is $A = \pi R^2$

$$\text{So, } \frac{dA}{dt} = 2\pi R \frac{dR}{dt}$$

The flux is,

$$\phi = 2\pi R \frac{dR}{dt} \times B$$

So, the magnitude of emf is

$$\begin{aligned}\epsilon &= 2 \times \frac{22}{7} \times 2 \times 10^{-2} \times 10^{-3} \times 0.4 \\ &= 5.028 \times 10^{-5} \text{ V} = 50.28 \mu\text{V} \approx 50 \mu\text{V}\end{aligned}$$

12. (2)

An electric field varying linearly with time will not produce variable magnetic field as current will be constant.

A decelerating charged particle can result in a time varying magnetic field by emitting radiation.

Similarly a permanent magnet has a constant magnetic field and a direct current does not change its direction and cannot produce a magnetic field.

An antenna can modulate the digital signals.

So, the right option is the statement D.

13. (25)

The formula to calculate the current flowing through an L-R circuit is given by

$$i = \frac{E}{R} \left[1 - e^{-\frac{Rt}{L}} \right] \dots (1)$$

The energy stored within the inductor can be calculated as follows:

$$\begin{aligned} E_{\text{induction}} &= \frac{1}{2} Li^2 \\ &= \frac{1}{2} L \frac{E^2}{R^2} \left[1 - e^{-\frac{Rt}{L}} \right]^2 \dots (2) \end{aligned}$$

The rate of change of energy stored within the inductor is given by

$$\begin{aligned} \frac{d}{dt} (E_{\text{induction}}) &= 2 \left[1 - e^{-\frac{Rt}{L}} \right] \cdot e^{-\frac{Rt}{L}} \cdot \frac{R}{L} \cdot \frac{E^2 L}{2R^2} \\ &= \frac{E^2}{R} \left[e^{-\frac{Rt}{L}} - e^{-\frac{2Rt}{L}} \right] \dots (3) \end{aligned}$$

For maximum rate,

$$\frac{d^2 E}{dt^2} = 0 \dots (4)$$

From equation (3) and (4), it can be written that

$$\begin{aligned} \frac{d}{dt} \left[\frac{E^2}{R} \left[e^{-\frac{Rt}{L}} - e^{-\frac{2Rt}{L}} \right] \right] &= 0 \\ \Rightarrow e^{-\frac{Rt}{L}} &= 2e^{-\frac{2Rt}{L}} \\ \Rightarrow e^{\frac{Rt}{L}} &= 2 \dots (5) \end{aligned}$$

From equation (3) and (5), it can be written, under maximum energy stored condition for the circuit, that

$$\begin{aligned} \frac{d}{dt} (E_{\text{induction}}) &= \frac{E^2}{R} [2^{-1} - 2^{-2}] \\ &= \frac{E^2}{4R} \end{aligned}$$

Substitute the values of the resistance and compare with the given expression to obtain the values of the parameters.

$$\begin{aligned} a &= 2 \\ b &= 2R \\ &= 50 \end{aligned}$$

Hence, the required ratio is given by

$$\frac{b}{a} = 25$$

14. (60)

The induced emf (ϵ) in the turns of the coil can be written as

$$\begin{aligned} \epsilon &= -\frac{d\phi}{dt} \\ &= iR \dots (1) \end{aligned}$$

Simplify equation (1) to obtain the required charge through a point in the circuit.

$$\begin{aligned} -\frac{d\phi}{dt} &= iR \\ \Rightarrow -\frac{1}{R} \int_{\phi_1}^{\phi_2} d\phi &= \int i dt \\ \Rightarrow q &= \frac{\phi_1 - \phi_2}{R} \dots (2) \end{aligned}$$

Where, ϕ_1 , ϕ_2 are the magnetic flux associated with the coil from two ends.

As, $\phi = NBA$, from equation (2), it can be written that

$$q = \frac{NA}{R} (B_1 - B_2) \dots (3)$$

where, N , A are the number of turns and the cross-sectional area of the core respectively.

Substitute the values of the known parameters into equation (3) to calculate the required charge.

$$\begin{aligned} q &= \frac{100 \times 24 \times 10^{-4} \text{ m}^2 \times (1.5 \text{ T} - (-1.5 \text{ T}))}{12 \Omega} \\ &= 0.06 \text{ C} \times \frac{1000 \text{ mC}}{1 \text{ C}} \\ &= 60 \text{ mC} \end{aligned}$$

15. (1)

The current (i) through the circuit is given by

$$i = \frac{V}{R}$$

$$= \frac{12 \text{ V}}{6 \Omega}$$

$$= 2 \text{ A}$$

The formula to calculate the induced emf in the coil can be written as

$$\varepsilon = L \frac{di}{dt} \dots (1)$$

Substitute the values of the known parameters into equation (1) and solve to calculate the inductance of the coil.

$$20 \text{ V} = L \times \frac{2 \text{ A}}{1 \times 10^{-3} \text{ s}}$$

$$\Rightarrow L = \frac{20 \text{ V} \times 1 \times 10^{-3} \text{ s}}{2 \text{ A}}$$

$$= 10 \text{ mH}$$

16. (88)

The data given is

$$\omega = 210 \times \frac{2\pi}{60} \text{ rad s}^{-1} = 7\pi \text{ rad s}^{-1}$$

$$l = 0.2 \text{ m}$$

$$B = 0.2 \text{ T}$$

The emf is given by

$$\varepsilon = \frac{B\omega l^2}{2}$$

$$= 0.2 \times 7\pi \times \frac{(0.2)^2}{2}$$

$$= 88 \text{ mV}$$